Kusner's Conjecture in ℓ_1^5

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Abstract

We prove Kusner's conjecture in ℓ_1^5: every equilateral subset of real five-dimensional Manhattan space is finite and has at most ten elements, and the bound is attained. The argument combines exact energy identities for ordered coordinate cuts with geometric width bounds on four-point subsets to exclude eleven equilateral points. A Lean formalization verifies the theorem for arbitrary real coordinates and every positive common distance.

1 Introduction and statement

For $x, y \in \mathbb{R}^d$, write

$$\operatorname{dist}_1(x, y) = \|x - y\|_1 = \sum_{k=1}^{d} |x_k - y_k| .$$

A set is equilateral if there is a real number $\delta > 0$ such that $\operatorname{dist}_1(x, y) = \delta$ whenever x and y are distinct members. The equilateral number $e(\ell_1^d)$ is the largest possible cardinality of such a set. The standard coordinate construction gives $e(\ell_1^d) \geq 2d$; the assertion that equality always holds is usually called Kusner's conjecture. Equilateral sets in finite-dimensional normed spaces and the special features of ℓ_p spaces are discussed in [4, 1]. Koolen, Laurent and Schrijver established the conjectured value in dimension four [4]. The recent account [3] still records the rectilinear equality for dimensions at most four. We prove the equality in dimension five.

Theorem 1.1. *Let $S \subseteq \mathbb{R}^5$ and let $\delta > 0$. If*

$$\sum_{k=1}^{5} |x_k - y_k| = \delta \qquad (x, y \in S,\ x \neq y),$$

then S is finite and $|S| \leq 10$. There is a unit-equilateral set of ten points in $\mathbb{R}^5$. Consequently $e(\ell_1^5) = 10$.

We prove the exclusion of eleven unit-equilateral points first. All coordinates in the argument are the five original scalar coordinates. Auxiliary clamps and projections are introduced explicitly, and their changes to distance are accounted for. Scalar ties and constant coordinates are allowed throughout.

The proof has two interacting parts. Ordered scalar cuts give exact trace and energy identities, which force at least four coordinate ranges to exceed one half and restrict their extreme pairs to five possible forests. Four of these possibilities are excluded by explicit prefix profiles, leaving a matching.

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To exclude the remaining matching, we prove that any four original points have total coordinate width at least 37/24. This bound uses the same seven remaining original points, signed overlaps, projections with exactly accounted displacement, and physical perimeters of plane sections of boxes. Applied to a suitable four-point subset in the matching, it couples the five coordinate ranges and forces positive slack away from the otherwise possible endpoint of the energy inequalities. The resulting contradiction has an explicit positive rational margin.

2 Orders, cuts, and exact energy identities

Suppose, for a contradiction, that $x_1, \dots, x_{11} \in \mathbb{R}^5$ satisfy

$$\operatorname{dist}_1(x_i, x_j) = 1 \qquad (i \neq j). \tag{1}$$

For coordinate k, choose a permutation p_k of the original labels such that

$$x_{p_k(1),k} \leq \cdots \leq x_{p_k(11),k}.$$

Any consistent order of tied entries may be used. Put

$$g_{k,r} = x_{p_k(r+1),k} - x_{p_k(r),k} \geq 0, \quad A_{k,r} = \{p_k(1), \dots, p_k(r)\}, \quad L_k = \sum_{r=1}^{10} g_{k,r}.$$

Thus L_k is the actual maximum minus minimum of coordinate k.

Let $J = \mathbf{1}\mathbf{1}^{\mathsf{T}}$ and $P = I - J/11$, acting on $\mathbb{R}^{11}$. Let D_k be the matrix with entries $|x_{i,k} - x_{j,k}|$, and define

$$C_k = -\frac{1}{2} P D_k P.$$

If χ_A is the indicator vector of a subset A, its cut matrix has entries 1 precisely when one label is in A and the other is not. This matrix is

$$\chi_A \mathbf{1}^{\mathsf{T}} + \mathbf{1}\chi_A^{\mathsf{T}} - 2\chi_A \chi_A^{\mathsf{T}}.$$

The scalar distance between two ordered entries is the sum of the gaps whose cuts separate them. Since $P\mathbf{1} = 0$, it follows that

$$C_k = \sum_{r=1}^{10} g_{k,r} (P\chi_{A_{k,r}})(P\chi_{A_{k,r}})^{\mathsf{T}} \succeq 0. \tag{2}$$

The complete distance equations give $\sum_k D_k = J - I$ and therefore

$$\sum_{k=1}^{5} C_k = \frac{1}{2} P, \qquad 0 \preceq C_k \preceq \frac{1}{2} P, \qquad \sum_{k=1}^{5} \operatorname{tr} C_k = 5. \tag{3}$$

Writing

$$w_r = \frac{r(11-r)}{11},$$

we also obtain the exact trace formula

$$T_k := \operatorname{tr} C_k = \sum_{r=1}^{10} g_{k,r} w_r, \tag{4}$$

because $\|P\chi_A\|_2^2 = |A|\,(11 - |A|)/11$.

For later use, take any real heights $h_0, \dots, h_{11}$ with $h_0 = h_{11} = 0$ and transport their increments to the original labels:

$$v_k(p_k(r)) = h_r - h_{r-1} \qquad (1 \le r \le 11).$$

Then $\sum_i v_k(i) = 0$, and its sum on $A_{k,r}$ equals h_r. Thus

$$v_k^{\mathsf{T}} C_k v_k = \sum_{r=1}^{10} g_{k,r} h_r^2, \qquad v_k^{\mathsf{T}} \Big(\sum_{\ell=1}^{5} C_\ell \Big) v_k = \frac{1}{2} \sum_{r=1}^{11} (h_r - h_{r-1})^2. \tag{5}$$

For a fixed profile put $a(h) = \frac{1}{2} \sum_r (h_r - h_{r-1})^2$. Its directed cross energy is

$$\Delta(h) := \sum_{k=1}^{5} \sum_{\ell \neq k} v_k^{\mathsf{T}} C_\ell v_k = 5a(h) - \sum_{k=1}^{5} \sum_{r=1}^{10} g_{k,r} h_r^2. \tag{6}$$

The sum is over ordered pairs of coordinates. Each summand is nonnegative. A constant scalar coordinate still has an order and a transported vector with the same value of $a(h)$; its own energy is zero. Hence constant columns are included in these identities without omission.

3 Triple geometry and the coordinate range bound

Lemma 3.1. *For three distinct original points a, b, c, their coordinatewise median m satisfies $\mathrm{dist}_1(a, m) = \mathrm{dist}_1(b, m) = \mathrm{dist}_1(c, m) = 1/2$. Their total coordinate width is $3/2$.*

Proof. In each scalar coordinate, the median lies on all three pairwise intervals. Consequently $\mathrm{dist}_1(a, b) = \mathrm{dist}_1(a, m) + \mathrm{dist}_1(b, m)$, and the other two analogous identities hold. The three left sides are one, so each of the three distances to m is one half. In one coordinate the sum of the three pairwise absolute differences is twice its maximum minus minimum. Summing over coordinates gives total width $3/2$. □

Call a coordinate *wide* if $L_k > 1/2$, and put $t_k = L_k - 1/2$ for a wide coordinate. Choose its minimum and maximum labels a_k, b_k. Applying Lemma 3.1 to these two points and any other label j, the median in coordinate k is $x_{j,k}$. Hence

$$x_{b_k,k} - \frac{1}{2} \le x_{j,k} \le x_{a_k,k} + \frac{1}{2} \qquad (j \notin \{a_k, b_k\}). \tag{7}$$

These bounds show that the two extreme labels are unique and that both outer ordered gaps are at least t_k. They also show that the width of any subset avoiding both extreme labels is at most $1 - L_k$. For every coordinate, wide or not, we therefore have

$$L_k - \frac{g_{k,1} + g_{k,10}}{2} \le \frac{1}{2}. \tag{8}$$

Indeed, in a wide column each outer gap is at least $L_k - 1/2$; in a nonwide column the gaps are nonnegative.

We require the following sharper universal range bound.

Proposition 3.2. *Every original coordinate satisfies $L_k \le 5/6$. Thus every wide excess satisfies $0 < t_k \le 1/3$.*

Proof. Only a wide coordinate needs consideration. Translate it so that its extreme values are $0, L$, and let a, b be their labels. In the remaining four coordinates, translate and reflect separately so that a is the zero vector and $b = (\ell_1, \dots, \ell_4)$, where

$$\ell_i \geq 0, \qquad \rho := \sum_{i=1}^{4} \ell_i = 1 - L.$$

For each of the nine other original points, let u_j denote its selected scalar coordinate. Clamp its remaining coordinates to the box $\prod_i [0, \ell_i]$, obtaining y_j, and write its remainder as w_j. The coordinatewise median of a, b, x_j is (u_j, y_j). Lemma 3.1 gives

$$\|w_j\|_1 = \frac{1}{2}, \qquad \sum_i y_{j,i} = \frac{1}{2} - u_j. \tag{9}$$

For real numbers define their signed overlap by

$$\omega(s,t) = \frac{|s| + |t| - |s - t|}{2}.$$

It is the smaller absolute value when s, t have the same sign, and zero when their signs are opposite. Set $I(v, w) = \sum_i \omega(v_i, w_i)$. Then

$$\|v - w\|_1 = \|v\|_1 + \|w\|_1 - 2I(v, w). \tag{10}$$

Both a scalar clamp and its remainder are monotone, and their increments sum to the original increment. Thus clamping splits every original distance exactly into the clamp distance and the remainder distance. Using (9) for two distinct outside labels yields

$$I(w_j, w_h) = \frac{1}{2}\left(\|y_j - y_h\|_1 + \left|\sum_i (y_{j,i} - y_{h,i})\right|\right) \leq \|y_j - y_h\|_1 . \tag{11}$$

If w_j, w_h have the same nonzero signed ray in coordinate i, their clamps have the same endpoint there. Therefore, for any set E of such coordinates,

$$I(w_j, w_h) + \sum_{i \in E} \ell_i \leq \rho. \tag{12}$$

Choose one label attaining the maximum remainder magnitude on each of the eight signed coordinate rays. There are nine labels, so some label j was not chosen on any ray. For each nonzero coordinate of w_j, assign its ray's chosen label as owner. Group the coordinates by their *distinct* owners; call their set H, and let E_h be the group assigned to h. We have $h \neq j$, $|H| \leq 4$, and

$$\sum_{i \in E_h} |w_{j,i}| \leq I(w_j, w_h).$$

This uses actual maximum magnitudes on the same rays, so each assigned coordinate contributes its full magnitude to the overlap. Summing (12) once for each distinct owner gives

$$\frac{1}{2} + \sum_{i \in \operatorname{supp} w_j} \ell_i \leq |H|\, \rho. \tag{13}$$

If $|H| \leq 3$, this implies $1/2 \leq 3\rho$. If $|H| = 4$, all four physical coordinates must belong to the support, so the width sum on the left is ρ and the same conclusion follows. Thus $1/2 \leq 3(1 - L)$, giving $L \leq 5/6$. Zero box widths cause no difficulty: their remainders and their signed rays were retained, and their contributions to the subtracted width sum are simply zero. □

4 Prefix profiles and wide-coordinate forests

We now evaluate four particular profiles. Their constants will also be used in the later matching estimates.

4.1 Forcing at least four wide coordinates

First take the symmetric profile whose first five positive heights are

$$(h_1, h_2, h_3, h_4, h_5) = \frac{1}{\sqrt{22}}(3, \sqrt{14}, \sqrt{26}, \sqrt{34}, \sqrt{38}), \qquad h_{11-r} = h_r.$$

Its squares equal $w_r - 1$ at the eight internal cuts and $w_r - 1/2$ at the two outer cuts. Expanding its increment squares gives

$$a_M = \frac{204 - 6\sqrt{14} - 4\sqrt{91} - 4\sqrt{221} - 4\sqrt{323}}{22}.$$

Equations (5) and (8) imply

$$T_k \le a_M + L_k - \frac{g_{k,1} + g_{k,10}}{2} \le \beta_M := a_M + \frac{1}{2} < \frac{21}{20}. \tag{14}$$

For a nonwide coordinate use instead $h_r = \sqrt{w_r - 2/3}$, with zero endpoint heights. Its first five heights are $\frac{1}{\sqrt{33}}(\sqrt{8}, \sqrt{32}, \sqrt{50}, \sqrt{62}, \sqrt{68})$. Its half squared increment norm is

$$a_N = \frac{260 - 20\sqrt{31} - 4\sqrt{1054}}{33}.$$

Thus

$$T_k \le a_N + \frac{2}{3}L_k \le a_N + \frac{1}{3} < \frac{10}{11} \qquad (L_k \le 1/2). \tag{15}$$

All strict numerical inequalities used here and below are certified by rational square comparisons in Section 10. If there were at most three wide coordinates, then

$$5 = \sum_k T_k < 3\frac{21}{20} + 2\frac{10}{11} = \frac{1093}{220} < 5,$$

where replacing a nonwide cap by a wide cap only increases the right side. This contradiction proves that at least four coordinates are wide.

4.2 The shifted and unshifted root profiles

For the remainder of the energy argument use

$$h_r = \sqrt{w_r - \frac{1}{2}}, \qquad h_0 = h_{11} = 0.$$

The first five heights are $\frac{1}{\sqrt{22}}(3, 5, \sqrt{37}, 3\sqrt{5}, 7)$, followed by their reflection. The squared increments, in positional order, are

$$s_1, s_2, s_3, s_4, s_5, 0, s_5, s_4, s_3, s_2, s_1, \tag{16}$$

where

$$s_1 = \frac{9}{22}, \qquad s_2 = \frac{2}{11}, \qquad s_3 = \frac{62 - 10\sqrt{37}}{22},$$

$$s_4 = \frac{82 - 6\sqrt{185}}{22}, \qquad s_5 = \frac{94 - 42\sqrt{5}}{22}. \tag{17}$$

They satisfy $s_1 > s_2 > s_3 > s_4 > s_5 > 0$. Put

$$\alpha = \sum_{i=1}^{5} s_i = \frac{251 - 10\sqrt{37} - 6\sqrt{185} - 42\sqrt{5}}{22} < \frac{2}{3}. \tag{18}$$

The own-coordinate energy is $T_k - L_k/2$. Writing $W = \sum_k L_k$, the exact directed cross energy is consequently

$$\Delta = 5\alpha + \frac{W}{2} - 5 \geq 0. \tag{19}$$

Also $T_k \leq \alpha + L_k/2$, so $W \geq 10 - 10\alpha > 10/3$. If

$$H = \sum_{k \text{ wide}} t_k,$$

then $W \leq 5/2 + H$, whence

$$H > \frac{5}{6}. \tag{20}$$

The unshifted root profile $h_r = \sqrt{w_r}$ has half squared increment norm

$$\alpha_0 = \frac{190 - 12\sqrt{5} - 24\sqrt{3} - 8\sqrt{42} - 4\sqrt{210}}{11} < \frac{43}{40}. \tag{21}$$

Its own energy is T_k, and its outer increment squares are $10/11$. Its total directed cross energy is exactly $5\alpha_0 - 5$.

4.3 The forest and its isolated edges

Associate to each wide coordinate the edge $\{a_k, b_k\}$ on the eleven original labels. No two coordinates have the same extreme pair: their two ranges alone would give distance greater than one for that pair. There is no cycle either. If a cycle uses m wide edges, the variation of each associated scalar coordinate along the entire cycle is at least twice its range. Summing those m coordinate variations gives more than m, although the cycle has m original unit-distance edges. This is impossible. Thus the wide-edge graph is a simple forest.

For every zero-sum vector v, its energy in a wide target coordinate is bounded below by its two outer cuts:

$$v^{\mathsf{T}} C_\ell v \geq t_\ell (v(a_\ell)^2 + v(b_\ell)^2). \tag{22}$$

Here the last prefix has sum $-v(b_\ell)$, explaining why both endpoint squares occur with the same factor t_ℓ. Apply this to the unshifted root vectors. An endpoint common to the source row and target edge has square $10/11$. If $\deg(a)$ denotes degree in the wide forest, summing over the ordered source-target pairs gives

$$5\alpha_0 - 5 \geq \frac{10}{11} \sum_{\ell \text{ wide}} t_\ell (\deg(a_\ell) + \deg(b_\ell) - 2). \tag{23}$$

For a nonisolated edge the degree factor is at least one; for an isolated edge it is zero. Hence the excess sum H_{non} of the nonisolated edges satisfies

$$H_{\text{non}} \leq \frac{11}{10}(5\alpha_0 - 5) < \frac{33}{80}.$$

Together with (20), this gives

$$H_{\text{iso}} > \frac{5}{6} - \frac{33}{80} = \frac{101}{240}. \tag{24}$$

Each individual excess is at most $1/3 = 80/240$ by Proposition 3.2; thus there are at least two isolated edges.

There are four or five wide edges. A nonisolated edge has a nonisolated neighbor, so there cannot be exactly one nonisolated edge. The complete list of possibilities is therefore

$$(m, i) = (4, 2), (4, 4), (5, 2), (5, 3), (5, 5), \tag{25}$$

where m is the number of wide edges and i the number of isolated ones. In $(4, 2)$ and $(5, 3)$ the nonisolated part is a two-edge path. In $(5, 2)$ it is a three-edge tree: it must be connected, since two nontrivial components would require at least four nonisolated edges; it then has four vertices because it is a forest. The cases $(4, 4)$ and $(5, 5)$ are matchings.

5 Excluding four of the five forest shapes

We use the shifted profile of (16). Its actual extreme entries have square $s_1 = 9/22$. The same counting as in (23) consequently yields

$$\Delta \geq \frac{9}{22} H_{\text{non}}. \tag{26}$$

No assertion about entries at other labels is needed for this inequality.

5.1 Two isolated edges

Suppose first that $(m, i) = (4, 2)$. The two isolated excesses sum to at most $2/3$, so $H_{\text{non}} \geq H - 2/3 \geq W - 19/6$. The four wide ranges and the nonwide range satisfy $W \leq 4(5/6) + 1/2 = 23/6$. Therefore

$$\Delta \geq \frac{9}{22}\left(W - \frac{19}{6}\right), \qquad \Delta < \frac{W}{2} - \frac{5}{3}$$

by (18) and (19). The lower bound minus the strict upper bound is

$$\frac{49}{132} - \frac{W}{11} \geq \frac{1}{44} > 0,$$

a contradiction.

Now suppose $(m, i) = (5, 2)$. The three-edge tree has four vertices and the two isolated edges have four more, leaving three original labels that are extreme in no coordinate. Call them core labels. Their total coordinate width is $3/2$ by Lemma 3.1. In each coordinate (7) bounds their width by $1 - L_k$, so

$$\frac{3}{2} \leq 5 - W, \qquad W \leq \frac{7}{2}, \qquad H = W - \frac{5}{2} \leq 1.$$

Again the isolated excesses sum to at most $2/3$. Thus

$$\Delta \geq \frac{9}{22}\left(H - \frac{2}{3}\right), \qquad \Delta < \frac{H}{2} - \frac{5}{12}.$$

Their difference is $19/132 - H/11 \geq 7/132 > 0$. This excludes $(5, 2)$.

5.2 A finite positional estimate

Let $q(r)$ be the squared increment at position r in (16). The smallest $2, 4, 6, 8$ entries of its eleven-entry multiset have respective sums

$$M_2 = s_5, \quad M_4 = 2s_5 + s_4, \quad M_6 = 2s_5 + 2s_4 + s_3, \quad M_8 = 2s_5 + 2s_4 + 2s_3 + s_2. \tag{27}$$

For any $2r$ distinct original labels their squared entries in a given row sum to at least M_{2r}. Positions remain distinct even when their scalar values are tied.

Define

$$\kappa_1 = s_5, \quad \kappa_2 = s_5 + s_4, \quad \kappa_3 = s_4 + s_3, \quad \kappa_4 = s_3 + s_2. \tag{28}$$

Suppose $u_1 \geq \cdots \geq u_r \geq 0$, where $r = 3$ or 4, and each u_j is assigned to two distinct labels, all $2r$ labels being distinct. If Q_j is the sum of their two squared entries, then

$$\sum_{j=1}^{r} u_j Q_j \geq \sum_{j=1}^{r} \kappa_j u_j. \tag{29}$$

Indeed, write the left side as

$$u_r \sum_{j=1}^{r} Q_j + \sum_{a=1}^{r-1} (u_a - u_{a+1}) \sum_{j=1}^{a} Q_j.$$

Every partial sum contains $2a$ distinct positions, so it is at least M_{2a}. Substitution of (27) proves (29). This argument is a lower relaxation of actual indexed positions; it never chooses new scalar orders or assumes that independently favorable placements can be realized.

We shall also need a weighted threshold version. For distinct positions $f(1), \ldots, f(n)$, nonnegative weights $u_j \leq M$, and $a \geq 0$,

$$\sum_j u_j q(f(j)) \geq a \sum_j u_j - M \sum_{r=1}^{11} (a - q(r))_+ . \tag{30}$$

To prove it, write

$$u_j q(f(j)) = a u_j - u_j (a - q(f(j))).$$

Bound the subtracted term by $M\,(a - q(f(j)))_+$, and use distinctness to bound the sum of these nonnegative discounts by the full positional sum. If one specified weight equals $S \geq M$ and all other weights are at most M, the same proof, separating the extra weight $S - M$, gives

$$\sum_j u_j q(f(j)) \geq a \sum_j u_j - M \sum_{r=1}^{11} (a - q(r))_+ - (S - M)a. \tag{31}$$

The last term is valid since $q(r) \geq 0$, so every discount is at most a. Directly from (16),

$$\sum_r (s_3 - q(r))_+ = 5s_3 - 2s_4 - 2s_5 =: D_3, \quad \sum_r (s_4 - q(r))_+ = 3s_4 - 2s_5 =: D_4. \tag{32}$$

5.3 The matching of four wide edges

Suppose $(m, i) = (4, 4)$, and order its wide excesses as $t_1 \geq t_2 \geq t_3 \geq t_4 > 0$. Write $L_0 \leq 1/2$ for the remaining range. The eight extreme labels leave three core labels. Their width in a wide coordinate is at most $1 - L_k$ and in the other coordinate at most L_0. Lemma 3.1 therefore gives

$$\frac{3}{2} \leq \sum_{k=1}^{4} (1 - L_k) + L_0 = 2 - H + L_0, \qquad H := \sum_{k=1}^{4} t_k \leq 1.$$

In each wide source row the three other edges contribute their two distinct endpoint positions through (22). Applying (29) with $r = 3$ gives the three coefficients $\kappa_1, \kappa_2, \kappa_3$ in descending weight order after omitting that row's own edge. In the nonwide source row all four edges contribute, and the coefficients are $\kappa_1, \ldots, \kappa_4$. Summing these five actual rows gives

$$\begin{aligned} &B_1 = 4\kappa_1, \qquad B_2 = \kappa_1 + 3\kappa_2, \qquad B_3 = 2\kappa_2 + 2\kappa_3, \\ &B_4 = 3\kappa_3 + \kappa_4, \qquad B_5 = 4\kappa_4, \end{aligned} \tag{33}$$

and

$$\Delta \geq \sum_{j=1}^{4} B_j t_j. \tag{34}$$

For example, t_2 receives κ_1 from the first wide row, κ_2 from the third and fourth wide rows, and another κ_2 from the nonwide row. This explains the ordered-row count and its factor three. Every edge weight has already been attached to its two original endpoint labels in (29).

The coefficients satisfy

$$B_1 < B_2 < B_3 < B_4 < B_5, \qquad B_3 < \frac{17}{64}, \qquad B_4 < \frac{1}{2}, \qquad B_5 + B_3 > 1. \tag{35}$$

Their exact certificates are proved in Section 10. Since $t_1, t_2 \leq 1/3$ and $t_4 \geq 0$, increasing coefficients give

$$\sum_{j=1}^{4} B_j t_j \geq \frac{B_1 + B_2}{3} + B_3 \left(H - \frac{2}{3} \right). \tag{36}$$

For clarity, the difference between the two sides is

$$(B_3 - B_1)(1/3 - t_1) + (B_3 - B_2)(1/3 - t_2) + (B_4 - B_3)t_4 \geq 0.$$

On the other hand $W = 2 + H + L_0 \leq 5/2 + H$, so

$$\Delta \leq 5\alpha - \frac{15}{4} + \frac{H}{2}. \tag{37}$$

Because $B_3 < 1/2$ and $H \leq 1$, the difference between the lower bound in (36) and the upper bound in (37) is at least

$$G_1 := \frac{B_1 + B_2 + B_3}{3} - 5\alpha + \frac{13}{4} = \frac{-3825 + 260\sqrt{37} + 96\sqrt{185} + 420\sqrt{5}}{132} > \frac{17549}{1650000} > 0. \tag{38}$$

This excludes the four-edge matching.

5.4 Three isolated edges and a two-edge path

It remains in this section to exclude $(m, i) = (5, 3)$. Let the three isolated excesses be u_1, u_2, u_3, with sum $U \leq 1$. Write the path as A–B–C, with excesses a, c and sum $S = a + c \leq 2/3$. There are nine distinct endpoint labels: six isolated endpoints and the three path vertices. Each wide source row has seven other endpoint labels in distinct internal positions.

Consider the source row belonging to edge AB. In the target edge BC, the shared label B is one of the source row's own extremes, so its square is s_1. Its contribution is cs_1. The other seven labels have weights $u_1, u_1, u_2, u_2, u_3, u_3, c$, each at most $1/3$. Equation (30) with

threshold s_3 bounds their contribution below by $s_3(2U+c)-D_3/3$. The analogous statement for row BC has as_1 and $s_3(2U+a)-D_3/3$. The two path rows together therefore contribute at least

$$4s_3U + (s_1+s_3)S - \frac{2}{3}D_3. \tag{39}$$

In the row of an isolated edge with excess u, the seven other labels have total weight $2(U-u)+2S$. The shared path vertex B occurs *once*, with weight $S=a+c$; the weights at A, C are a, c. The other four weights are the endpoints of the two other isolated edges. If $S \le 1/3$, all seven weights are at most $1/3$, and the threshold-s_4 discount is at most $D_4/3$. If $S \ge 1/3$, only the single shared weight can exceed $1/3$. Equation (31) bounds the discount by

$$\frac{D_4}{3} + (S-1/3)s_4 = s_4S + \frac{2}{3}(s_4-s_5).$$

This bound holds wherever the shared label is actually positioned.

Summing these three isolated rows with (39) gives the following lower bounds for the same original Δ:

$$F_{\text{small}}(U,S) = 4(s_3+s_4)U + (s_1+s_3+6s_4)S - \frac{5s_4+10s_3-10s_5}{3} \qquad (S \le 1/3), \tag{40}$$

$$F_{\text{large}}(U,S) = 4(s_3+s_4)U + (s_1+s_3+3s_4)S - \frac{2s_4+10s_3-10s_5}{3} \qquad (S \ge 1/3). \tag{41}$$

The bounds agree at $S=1/3$. Since all five coordinates are wide,

$$\Delta = 5\alpha - \frac{15}{4} + \frac{U+S}{2}. \tag{42}$$

The coefficient signs needed to compare these affine functions are

$$4(s_3+s_4) - \frac{1}{2} < 0, \qquad s_1+s_3+3s_4 - \frac{1}{2} > 0. \tag{43}$$

In the small-S case, replacing $U \le 1$ by 1 and $S \ge 0$ by zero can only decrease $F_{\text{small}} - \Delta$. Its resulting value is exactly G_1. In the large-S case, replacing U by 1 and S by $1/3$ can only decrease $F_{\text{large}} - \Delta$. Its value is

$$G_1 + \frac{1}{3}\left(s_1+s_3+6s_4-\frac{1}{2}\right) > G_1.$$

Both are positive by (38) and (43), contradicting $\Delta \ge F$. This excludes $(5,3)$.

Thus the only possible wide forest consists of five disjoint edges. Before excluding that matching, we establish a bound for every four-point subset using all seven of its remaining original labels.

6 Four original points and their seven remaining labels

Fix any four distinct original labels, denoted q_0, q_1, q_2, q_3. In each physical coordinate sort their four values. Its first and last gaps are singleton cuts, assigned respectively to the least and greatest label, and its middle gap is one of the three balanced two–two cuts. Let b_i be the total singleton width assigned to q_i, and let c_1, c_2, c_3 be the total widths of the three balanced cuts.

For complementary pairs, the balanced cuts make equal contributions. Subtracting their original unit-distance equations gives

$$b_0+b_1 = b_2+b_3, \quad b_0+b_2 = b_1+b_3, \quad b_0+b_3 = b_1+b_2.$$

Thus $b_0 = b_1 = b_2 = b_3 =: b \geq 0$. Each of the six pairs crosses two of the three balanced cuts. Their equations now imply

$$c_1 = c_2 = c_3 =: c = \frac{1}{2} - b \geq 0. \tag{44}$$

The actual total coordinate width W_Q of the four points is therefore

$$W_Q = 4b + 3c = \frac{3}{2} + b. \tag{45}$$

These identities also hold when a sorted gap is zero.

6.1 Clamps, signed owners, and all four cross equations

In each coordinate clamp every original point to the interval between the second and third selected values and subtract that interval's midpoint. Let its halfwidth be $\lambda_k \geq 0$. Reflect the coordinate if necessary and assign its balanced type $\tau(k) \in \{1, 2, 3\}$ so that the selected clamps are $H_{i,\tau(k)}\lambda_k$, where

$$H = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}, \qquad H^{\mathsf{T}}\mathbf{1} = 0, \quad H^{\mathsf{T}}H = 4I_3. \tag{46}$$

Every balanced partition of four labels is a column of this matrix, up to sign, so this uses only actual coordinate reflections. Coordinates with zero middle gap are still assigned a type. Write the resulting decomposition of each point as

$$x_j = y_j + w_j + \text{midpoint}, \qquad -\lambda_k \leq y_{j,k} \leq \lambda_k,$$

with the understood coordinate reflections on the left side. They preserve the physical ℓ_1 distance. As before, monotonicity of the clamp and its remainder gives the exact splitting

$$\operatorname{dist}_1(x_j, x_\ell) = \|y_j - y_\ell\|_1 + \|w_j - w_\ell\|_1 \quad \text{for every original pair.} \tag{47}$$

In particular a zero halfwidth forces $y_{j,k} = 0$ but does not force $w_{j,k} = 0$.

Let $u_i = w_{q_i}$ be the four selected tails. Their masses are $\|u_i\|_1 = b$. In one coordinate only the selected least value can have a negative tail and only the greatest can have a positive tail. Thus two owners never occupy the same nonzero signed ray; opposite rays of one coordinate are allowed. Also

$$\sum_{k:\tau(k)=t} \lambda_k = \frac{c}{2}, \qquad \Lambda := \sum_k \lambda_k = \frac{3c}{2} = \frac{3}{4} - \frac{3b}{2}. \tag{48}$$

The norm of each selected point relative to these midpoints is $b + \Lambda = 3/4 - b/2$. This quantity differs from Λ.

There are exactly seven other original labels. For each such label j define

$$A_{i,j} = I(u_i, w_j), \quad J_j = \sum_{i=0}^{3} A_{i,j}, \quad r_j = \|w_j\|_1, \quad Q_{t,j} = \sum_{k:\tau(k)=t} y_{j,k}.$$

The signed overlap of (10) satisfies

$$0 \leq A_{i,j} \leq b. \tag{49}$$

In a selected clamp coordinate, the distance to an arbitrary clamp is $\lambda_k - H_{i,\tau(k)} y_{j,k}$. The original pair (q_i, j) therefore gives, for each of the four values of i,

$$1 = b + \Lambda + r_j - (HQ_j)_i - 2A_{i,j}. \tag{50}$$

Summing these four equations, and then multiplying them by H^{T}, yields respectively

$$r_j = \delta + \frac{J_j}{2}, \qquad Q_j = -\frac{1}{2} H^{\mathsf{T}} A_j, \qquad \delta = \frac{1}{4} + \frac{b}{2}. \tag{51}$$

Here A_j is the four-entry agreement vector. Thus no equality of the original outsider norms has been assumed. Equations (51) are consequences of all four original cross distances.

6.2 Co-occurrence on physical signed rays

For two remaining labels j, ℓ the disjoint signed owner rays imply

$$I(w_j, w_\ell) \geq \sum_{i=0}^{3} (A_{i,j} + A_{i,\ell} - b)_+ . \tag{52}$$

Here is a coordinatewise proof that also accounts for opposite owner rays in a single coordinate. For nonnegative u, a, a',

$$\min(u, a, a') \geq \min(u, a) + \min(u, a') - u.$$

Equivalently, in signed-overlap notation, for any real u, a, a',

$$\min\{\omega(u, a), \omega(u, a')\} \geq \omega(u, a) + \omega(u, a') - |u| .$$

This follows because each overlap is at most $|u|$. Sum over coordinates for one owner u_i, and use nonnegativity of the left side to bound it below by $(A_{i,j} + A_{i,\ell} - \|u_i\|_1)_+$. Now sum over owners. If w_j, w_ℓ have opposite signs in a coordinate, all the minima in that coordinate are zero. If they have the same sign, only the one possible owner on that signed ray can contribute, and its minimum overlap is at most $\omega(w_{j,k}, w_{\ell,k})$. Thus no physical overlap is counted twice. This proves (52) using $\|u_i\|_1 = b$.

7 Projection with exactly accounted displacement

Lemma 7.1. *Let $\lambda_1, \ldots, \lambda_n \geq 0$ and $v \in \prod_i [-\lambda_i, \lambda_i]$. For every feasible sum $-\sum_i \lambda_i \leq T \leq \sum_i \lambda_i$ there exists a vector z in the same box such that*

$$\sum_i z_i = T, \qquad \sum_i |z_i - v_i| = \left| \sum_i v_i - T \right| .$$

In particular, the distance to the zero-sum section is exactly $|\sum_i v_i|$.

Proof. Write $Q = \sum_i v_i$ and $L = \sum_i \lambda_i$. If $Q = T$, take $z = v$. If $T < Q$, then $Q + L > 0$ and

$$\theta = \frac{Q - T}{Q + L} \in [0, 1], \qquad z_i = (1 - \theta) v_i - \theta \lambda_i$$

defines a point in the box with sum T. Each $z_i \leq v_i$, so the sum of absolute changes is $Q - T$. The case $Q < T$ follows by negating v, T, z. Finally every vector with sum T is at distance at least $|Q - T|$, by the triangle inequality for the sum of coordinate differences. This proves exactness, including zero widths. □

Lemma 7.2. *Assume in addition that $n \geq 2$ and all $\lambda_i > 0$. If v touches a box face, a closest projection to the zero-sum section can be chosen on that section's relative boundary, with the same displacement $|\sum_i v_i|$.*

Proof. After negation we may suppose $v_i = \lambda_i$ for some i. Put $M = \sum_{h \neq i} \lambda_h$. If $\lambda_i \leq M$, keep $z_i = \lambda_i$ and use Lemma 7.1 on the other coordinates with target sum $-\lambda_i$. Its displacement is exactly $|\sum_h v_h|$, and z still touches the ith face. If $\lambda_i > M$, take instead

$$z_i = M, \qquad z_h = -\lambda_h \quad (h \neq i).$$

This vector belongs to the box, has zero sum, and is coordinatewise at most v. Also $\sum_h v_h \geq \lambda_i - M > 0$, so its displacement is again $|\sum_h v_h|$. It touches every other face. Such another coordinate exists and has positive width. A saturated coordinate is a true relative-boundary condition here: the zero vector is strictly inside all the box inequalities, and each coordinate functional is nonconstant on the zero-sum subspace when $n \geq 2$. □

Apply Lemma 7.1 separately to the coordinates of each of the three actual balanced types. For each remaining original label j obtain z_j in the physical section

$$K = \left\{ z \in \mathbb{R}^5 : |z_k| \leq \lambda_k \ (1 \leq k \leq 5), \quad \sum_{k:\tau(k)=t} z_k = 0 \ (t = 1, 2, 3) \right\}. \tag{53}$$

Its displacement is exactly

$$E_j := \|z_j - y_j\|_1 = \sum_{t=1}^{3} |Q_{t,j}| = \frac{1}{2} \sum_{t=1}^{3} \left| (H^{\mathsf{T}} A_j)_t \right|. \tag{54}$$

Zero-width coordinates of z_j, y_j are zero; their original tails and agreements remain in (51) and (52).

7.1 The eight agreement forms and the pair table

For a real four-vector A let $J(A) = \sum_i A_i$, and put

$$E(A) = \frac{1}{2} \left(|A_0 + A_1 - A_2 - A_3| + |A_0 - A_1 + A_2 - A_3| + |A_0 - A_1 - A_2 + A_3| \right).$$

The following exact identity is useful:

$$F(A) := \frac{J(A)}{2} + E(A) = \max_{0 \leq i \leq 3} \{2A_i, \ J(A) - 2A_i\}. \tag{55}$$

To verify it, sort the four entries as $a \leq b' \leq c' \leq d$. The three absolute expressions are indexed by the three unordered balanced partitions and are therefore unchanged as a multiset by permuting the entries. Their half sum becomes

$$d - a + \frac{1}{2} \left| a + d - b' - c' \right|.$$

Adding half the total gives $\max(2d, a + b' + c' + d - 2a)$, precisely the right side of (55). This proof includes all ties.

Call $2A_i$ a plus form at index i and $J(A) - 2A_i$ a minus form at index i. Choose one maximizing form at each of the seven labels, once and for all. The coefficient vector of a plus form is 2 at its index and zero elsewhere. The coefficient vector of a minus form is -1 at its index and 1 elsewhere.

For $0 \le u, v \le b$ and real coefficients a, d, the function

$$au + dv - 2\,(u + v - b)_+$$

is linear on each triangle into which $u + v = b$ divides the square. Checking their vertices gives the upper bound

$$au + dv - 2\,(u + v - b)_+ \le b \max(0, a, d, a + d - 2). \tag{56}$$

For $b = 0$ the assertion is immediate. For $b > 0$, divide the variables by b; the four vertex values are $0, a, d, a + d - 2$, which proves it. Apply this independently to the four agreement slots and sum. For any two chosen forms the resulting total cap is b times the following number:

Chosen forms	Equal indices	Different indices
plus / plus	2	4
minus / minus	3	4
plus / minus	5	4

For example, equal-index plus/minus forms have cap 2 at their common index and cap 1 at each of the other three indices, totaling 5. Different-index plus/minus forms have cap 2 at the plus index, zero at the minus index, and 1 at each remaining index, totaling 4. For equal-index minus/minus forms the common-index cap is zero and the other three caps are 1. The other entries follow from the same four-vertex rule. Thus the table bounds actual forms, without any simultaneous-attainment assertion.

From (47), (51), the two projection displacements, and (52), we obtain

$$\begin{aligned} \|z_j - z_\ell\|_1 &\ge 1 - 2\delta - \left(F(A_j) + F(A_\ell) - 2 \sum_i (A_{i,j} + A_{i,\ell} - b)_+ \right) \\ &\ge \frac{1}{2} - b - b\,C(j, \ell), \end{aligned} \tag{57}$$

where $C(j, \ell)$ is the corresponding table entry. In particular,

$$\|z_j - z_\ell\|_1 \ge \frac{1}{2} - 6b \qquad (j \ne \ell). \tag{58}$$

To see the first line directly, the original distance equation gives $\|y_j - y_\ell\|_1 = 1 - r_j - r_\ell + 2I(w_j, w_\ell)$. Subtract $E_j + E_\ell$ by the triangle inequality and substitute $r_j = \delta + J_j/2$ and $E_j = E(A_j)$.

7.2 A seven-label cyclic lower bound

In any cycle on the same seven labels, there is an edge joining forms of the same sign, since signs cannot alternate around an odd cycle. If its indices are also equal, its table cap is at most 3; the other six caps are at most 5, giving total at most 33. If its indices differ, its cap is 4. There must be a second edge whose indices differ: otherwise equality of indices along the other six edges would force equality across the first edge too. The second cap is also 4, while the remaining five are at most 5. Again the total is at most 33. Summing (57) proves

$$\sum_{j \text{ in any cyclic order}} \|z_j - z_{j+1}\|_1 \ge \frac{7}{2} - 40b. \tag{59}$$

The forms are chosen per label, so the same choices work in whatever cyclic order is later supplied by geometry.

8 Physical boundary tours and the quartet width bound

We now prove

Proposition 8.1. *Every four-point subset of the eleven original points has $b \geq 1/24$ and total coordinate width $W_Q \geq 37/24$.*

Assume for a contradiction that $b < 1/24$. Then $c = 1/2 - b > 0$, and each balanced type has positive total halfwidth $c/2$. Let n_t be the number of its positive physical halfwidths. The type supports are disjoint and there are only five coordinates, so, after permuting the three types, the possibilities for (n_1, n_2, n_3) are

$$(1,1,1), \quad (2,1,1), \quad (3,1,1), \quad (2,2,1). \tag{60}$$

Every zero-width slot has $z_k = 0$ in the auxiliary section, irrespective of its original tail. The pair lower bound (58) now gives

$$\|z_j - z_\ell\|_1 > \frac{1}{4} \qquad (j \neq \ell). \tag{61}$$

For $(1,1,1)$ the section K is a point, contradicting (61). For $(2,1,1)$ its only nonzero coordinates have the form $(r, -r)$, with $|r| \leq A := \min(\lambda_1, \lambda_2)$. Their physical distance is $2\,|r - r'|$ and the segment diameter is $4A \leq c \leq 1/2$, since $\lambda_1 + \lambda_2 = c/2$. Three ordered points on a segment with successive distances greater than $1/4$ would have endpoint distance greater than $1/2$. Three of the seven labels already give a contradiction. No boundary condition was used in these two cases.

8.1 A genuine positive face for every remaining label

In each of the last two allocations in (60), all five physical coordinates have positive halfwidth. This fact must be established before passing from a tail to an intrinsic boundary. If a balanced type has a single coordinate, its halfwidth is $c/2$. Equation (51), together with (49), gives

$$|Q_{t,j}| \leq b < \frac{c}{2}.$$

Indeed each column of H has two plus and two minus entries, so the absolute difference of the corresponding two agreement sums is at most $2b$. Thus a singleton type cannot carry a nonzero original tail: such a tail would force its clamp to an endpoint, making $|Q_{t,j}| = c/2$.

On the other hand (51) gives $\|w_j\|_1 = \delta + J_j/2 > 0$. Hence some actual positive-width coordinate of a type with at least two coordinates has a nonzero tail. Its clamp touches a face. Apply Lemma 7.2 on the actual positive coordinates of that type, and use Lemma 7.1 on the other types. The resulting $z_j \in K$ still has exactly the displacement (54), and it touches a positive face of a nonsingleton section. The face witness is therefore inside the active type, rather than an outside zero coordinate. We use these choices for the seven labels.

8.2 The two paired coordinates in each of two types

For allocation $(2,2,1)$ the section, after an actual coordinate permutation, consists of

$$(r, -r, s, -s, 0), \qquad |r| \leq A, \quad |s| \leq B,$$

where each of A, B is the smaller of its type's two halfwidths. The physical metric on this rectangle is

$$2\,|r - r'| + 2\,|s - s'|\,. \tag{62}$$

Each selected z_j lies on a true side, by the positive-face conclusion above. Traversing the four literal vertices $(A, -A, B, -B, 0)$, $(-A, A, B, -B, 0)$, $(-A, A, -B, B, 0)$, $(A, -A, -B, B, 0)$ gives side lengths $4A, 4B, 4A, 4B$. Since $A, B \le c/4$, the complete tour length is

$$8(A + B) \le 4c = 2 - 4b. \tag{63}$$

The factors two in (62) come from two separate physical coordinates in each pair.

8.3 The three-coordinate plane section

For allocation $(3, 1, 1)$ the two singleton auxiliary coordinates are zero. The remaining section is the literal plane polygon

$$z_1 + z_2 + z_3 = 0, \qquad |z_1| \le a, \quad |z_2| \le d, \quad |z_3| \le f, \qquad a + d + f = \frac{c}{2}. \tag{64}$$

We give its physical boundary tour explicitly. On an edge with one coordinate fixed, the other two coordinate differences are opposites. Its physical distance is therefore *twice* the absolute change of either free coordinate.

If all three widths satisfy the triangle inequalities, use the six vertices, in order,

$$\begin{aligned} v_0 &= (a, f - a, -f), \quad v_1 = (a, -d, d - a), \\ v_2 &= (d - f, -d, f), \quad v_3 = (-a, a - f, f), \\ v_4 &= (-a, d, a - d), \quad v_5 = (f - d, d, -f). \end{aligned} \tag{65}$$

They satisfy all the box inequalities. Their six edge lengths are

$$2(d + f - a),\ 2(a + f - d),\ 2(a + d - f),\ 2(d + f - a),\ 2(a + f - d),\ 2(a + d - f). \tag{66}$$

Consequently the total physical length is $4(a + d + f) = 2c$. Every saturated box face is exactly its corresponding segment in this list. For example, on $z_1 = a$ the plane and box inequalities give $-d \le z_2 \le f - a$, the segment from v_1 to v_0. For completeness the six face intervals are

Fixed coordinate	Free coordinate	Its interval
$z_1 = a$	z_2	$[-d, f - a]$
$z_2 = -d$	z_1	$[d - f, a]$
$z_3 = f$	z_1	$[-a, d - f]$
$z_1 = -a$	z_2	$[a - f, d]$
$z_2 = d$	z_1	$[-a, f - d]$
$z_3 = -f$	z_1	$[f - d, a]$

The remaining coordinate is fixed by the plane equation. This verifies coverage of all true faces, including endpoint coincidences. If a triangle inequality is an equality, the corresponding edges in (66) have zero length and are retained as constant segments.

If a width dominates, permute the actual coordinates and suppose $a > d + f$. Then the plane implies $|z_1| \le d + f < a$; the first coordinate's original faces do not meet the section. The actual section is the rectangle with vertices

$$(d + f, -d, -f),\ (f - d, d, -f),\ (-d - f, d, f),\ (d - f, -d, f). \tag{67}$$

The four physical lengths are $4d, 4f, 4d, 4f$, summing to $8(d + f) \le 4(a + d + f)$. Every feasible saturated face is one of $z_2 = \pm d$ or $z_3 = \pm f$ and is covered by a displayed edge. This

proves for every choice of nonnegative widths a closed physical tour covering the saturated section faces with length at most

$$4(a + d + f) = 2c \le 4c = 2 - 4b. \tag{68}$$

The positive widths in the present allocation ensure that our selected face points are on its true boundary. The formulas themselves continue to cover zero and repeated-edge degeneracies.

8.4 Comparing the same seven-point cycle

A closed polygonal tour of physical length P admits a parametrization $\gamma : [0, P] \longrightarrow \mathbb{R}^5$ with $\gamma(0) = \gamma(P)$ and

$$\|\gamma(t) - \gamma(s)\|_1 \le t - s \qquad (0 \le s \le t \le P).$$

This follows by parametrizing each nonzero straight edge by its physical length and keeping zero edges constant. Choose a covering parameter for each of the seven z_j and sort these parameters. The distances between successive points are bounded by the successive parameter gaps; the closing distance is at most the sum of the two gaps through the common endpoint of the closed tour. Hence, in this particular cyclic order,

$$\sum_j \|z_j - z_{j+1}\|_1 \le P.$$

Repeated parameters cause no gap in this argument; in the present application they are also ruled out by (61).

Both allocations with five positive slots supply such a tour with $P \le 2 - 4b$, by (63) or (68). Apply the lower bound (59) to exactly this cyclic ordering of the same seven original labels. It gives

$$\frac{7}{2} - 40b \le 2 - 4b, \qquad b \ge \frac{1}{24},$$

contrary to our assumption. This completes the proof of Proposition 8.1, using (45) for its width conclusion.

9 The final five-edge matching

We have excluded all cases of (25) except $(5, 5)$. Relabel the five actual coordinates so that

$$L_i = \frac{1}{2} + t_i, \qquad \frac{1}{3} \ge t_1 \ge t_2 \ge t_3 \ge t_4 \ge t_5 > 0.$$

Their ten extreme labels a_i, b_i are all distinct, and let o be the eleventh label. Select the original quartet

$$Q = \{o, a_5, b_5, a_4\}.$$

Its width in coordinate five is L_5. In coordinate four it contains the actual minimum a_4, while the other three labels are nonextreme; (7) therefore bounds this coordinate width by $1/2$. In each of coordinates one, two, and three, all four labels avoid the two extremes, so their width is at most $1 - L_i$. Applying Proposition 8.1 to this same original quartet gives

$$\frac{37}{24} \le W_Q \le L_5 + \frac{1}{2} + \sum_{i=1}^{3}(1 - L_i) = \frac{7}{2} - W + 2L_5 + L_4. \tag{69}$$

Its seven outsiders are exactly the seven original labels used by the quartet proof; no new auxiliary equilateral configuration is introduced.

For the actual nonnegative cap slacks $d_i = 1/3 - t_i$, $1 \le i \le 4$, inequality (69) is equivalent to

$$d_1 + d_2 + d_3 + t_5 \ge \frac{1}{24}. \tag{70}$$

Indeed it says $t_1 + t_2 + t_3 - t_5 \le 23/24$. The fourth slack cancels in this conversion, and the fifth excess has the positive sign displayed in (70).

In each wide source row, the other four edges contribute eight distinct endpoint labels. Apply (29) to those four weights in their inherited descending order and sum the five rows. The result is

$$\Delta \ge F := \sum_{i=1}^{5} B_i t_i, \tag{71}$$

with exactly the coefficients of (33). To check the ordered count explicitly, the coefficient rows, with columns indexed by $t_1, \dots, t_5$, are

$$\begin{pmatrix} 0 & \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \\ \kappa_1 & 0 & \kappa_2 & \kappa_3 & \kappa_4 \\ \kappa_1 & \kappa_2 & 0 & \kappa_3 & \kappa_4 \\ \kappa_1 & \kappa_2 & \kappa_3 & 0 & \kappa_4 \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 & 0 \end{pmatrix}. \tag{72}$$

For example t_3 receives two copies of κ_2 and two of κ_3, and t_5 receives four copies of κ_4. The two endpoint occurrences per edge are already included in each κ through (27); no further factor is inserted.

All five coordinates are wide, so the exact same energy is

$$\Delta = 5\alpha - \frac{15}{4} + \frac{1}{2}\sum_{i=1}^{5} t_i.$$

Substitute $t_i = 1/3 - d_i$ for $i \le 4$ and subtract. This gives the identity

$$F - \Delta = G_4 + \sum_{i=1}^{4} \left(\frac{1}{2} - B_i\right) d_i + \left(B_5 - \frac{1}{2}\right) t_5, \tag{73}$$

where

$$G_4 := \frac{B_1 + B_2 + B_3 + B_4}{3} - 5\alpha + \frac{37}{12} = \frac{-2851 + 180\sqrt{37} + 60\sqrt{185} + 420\sqrt{5}}{132}. \tag{74}$$

The constant G_4 need not be positive. The original quartet coupling (70), rather than its sign alone, supplies the contradiction.

Put $\gamma = 1/2 - B_3$. By (35),

$$\gamma > \frac{15}{64}, \qquad \frac{1}{2} - B_i \ge \gamma \ (i = 1, 2, 3), \qquad \frac{1}{2} - B_4 > 0, \qquad B_5 - \frac{1}{2} > \gamma.$$

Since all the slacks and t_5 are nonnegative,

$$F - \Delta \ge G_4 + \gamma(d_1 + d_2 + d_3 + t_5) \ge G_4 + \frac{\gamma}{24}. \tag{75}$$

The exact radical certificate in Section 10 is $G_4 > -4349/660000$. Therefore

$$F - \Delta > -\frac{4349}{660000} + \frac{15}{64 \cdot 24} = -\frac{4349}{660000} + \frac{5}{512} = \frac{33541}{10560000} > 0. \tag{76}$$

This contradicts (71). The five-edge matching is impossible, and the complete forest classification proves that eleven original unit-equilateral points in $\mathbb{R}^5$ cannot exist.

10 Exact arithmetic certificates

This section supplies the numerical signs used in the proof. Every comparison concerns nonnegative real numbers and follows by squaring rational bounds. No decimal approximation is required.

10.1 The three profile bounds

For the modified profile, the following integers certify strict lower square-root bounds: in each row $m^2 < Nd^2$, hence $\sqrt{N} > m/d$.

N	m	d	$Nd^2 - m^2$
14	37416	10000	42944
91	95393	10000	175551
221	148660	10000	204400
323	179722	10000	2716

Their weighted numerator is

$$6(37416) + 4(95393) + 4(148660) + 4(179722) = 1919596.$$

It follows that

$$\beta_M = \frac{215 - 6\sqrt{14} - 4\sqrt{91} - 4\sqrt{221} - 4\sqrt{323}}{22} < \frac{57601}{55000} < \frac{21}{20},$$

as asserted in (14).

For the nonwide profile,

$$139^2 < 31 \cdot 25^2, \qquad 1623^2 < 1054 \cdot 50^2.$$

Thus $20\sqrt{31} + 4\sqrt{1054} > 6026/25$, and

$$a_N + \frac{1}{3} = \frac{271 - 20\sqrt{31} - 4\sqrt{1054}}{33} < \frac{749}{825} < \frac{10}{11}.$$

For the unshifted root profile, the following positive differences give the analogous lower bounds with denominator 1000:

N	m	$N \cdot 1000^2 - m^2$
5	2236	304
3	1732	176
42	6480	9600
210	14491	10919

The weighted sum is $12(2236) + 24(1732) + 8(6480) + 4(14491) = 178204$. Consequently

$$\alpha_0 < \frac{2949}{2750} < \frac{43}{40},$$

which proves (21) and the strict isolated-mass bound.

10.2 Positive increments and their upper and lower bounds

The definitions (17) also have the useful forms

$$s_3 = \frac{(\sqrt{37}-5)^2}{22}, \qquad s_4 = \frac{(3\sqrt{5}-\sqrt{37})^2}{22}, \qquad s_5 = \frac{(7-3\sqrt{5})^2}{22}. \tag{77}$$

The unsquared increments are positive. For the first, $\sqrt{37} > 6$. For the second, the rational square comparisons

$$\sqrt{5} > \frac{559}{250}, \qquad \sqrt{37} < \frac{6083}{1000}$$

give $3\sqrt{5} - \sqrt{37} > 5/8$. For the third, $\sqrt{5} < 7/3$. We may therefore square the following upper bounds without reversing a sign:

$$\sqrt{37} - 5 < \frac{11}{10}, \qquad 3\sqrt{5} - \sqrt{37} < \frac{3}{4}, \qquad 7 - 3\sqrt{5} < \frac{2}{5}.$$

They follow respectively from $\sqrt{37} < 61/10$, from $\sqrt{37} > 6$ and $\sqrt{5} < 9/4$, and from $\sqrt{5} > 11/5$. Equation (77) yields

$$s_3 < \frac{121}{2200} < \frac{1}{16}, \qquad s_4 < \frac{9}{352} < \frac{1}{32}, \qquad s_5 < \frac{2}{275} < \frac{1}{128}. \tag{78}$$

Similarly $\sqrt{37} > 121/20$ gives $\sqrt{37} - 5 > 21/20$, and the second increment is greater than $5/8$. Thus

$$s_3 > \frac{441}{8800} > \frac{1}{20}, \qquad s_4 > \frac{25}{1408} > \frac{1}{60}, \qquad s_5 > 0. \tag{79}$$

All auxiliary root comparisons above are checked by the following integer inequalities, in addition to $36 < 37$:

$$559^2 = 312481 < 5 \cdot 250^2 = 312500, \qquad 6083^2 = 37002889 > 37 \cdot 1000^2 = 37000000,$$
$$7^2 > 5 \cdot 3^2, \quad 61^2 > 37 \cdot 10^2, \quad 9^2 > 5 \cdot 4^2, \quad 11^2 < 5 \cdot 5^2, \quad 121^2 < 37 \cdot 20^2.$$

The upper and lower bounds imply $s_1 > s_2 > s_3 > s_4 > s_5 > 0$. They also prove (43): the first coefficient is less than $4(1/16 + 1/32) - 1/2 = -1/8$, and the second is greater than $9/22 + 1/20 + 3/60 - 1/2 = 1/110$.

10.3 The shifted trace bound and matching constants

The three root bounds needed for the remaining exact constants are

$$\sqrt{37} > \frac{608276}{100000}, \qquad \sqrt{185} > \frac{1360147}{100000}, \qquad \sqrt{5} > \frac{223606}{100000}. \tag{80}$$

Their square comparisons are

$$608276^2 = 369999692176 < 370000000000,$$
$$1360147^2 = 1849999861609 < 1850000000000,$$
$$223606^2 = 49999643236 < 50000000000.$$

For α, their weighted numerator is

$$10(608276)+6(1360147)+42(223606) = 23635094, \qquad 3(23635094) = 70905282 > 70900000.$$

Thus the subtracted radical sum in (18) exceeds $709/3$, giving $\alpha < 2/3$.

The successive differences of the coefficients in (33) are

$$\begin{aligned} B_2 - B_1 &= 3s_4, \\ B_3 - B_2 &= 2s_3 + s_4 - 2s_5, \\ B_4 - B_3 &= s_2 + 2s_3 - s_4 - 2s_5, \\ B_5 - B_4 &= 3(s_2 - s_4). \end{aligned}$$

They are all positive by the decreasing positive s_i sequence. Also (78) gives

$$B_3 = 2(s_3 + 2s_4 + s_5) < \frac{17}{64}, \qquad B_4 = s_2 + 4s_3 + 3s_4 < \frac{4211}{8800} < \frac{1}{2}.$$

Using (79),

$$B_5 + B_3 = 4s_2 + 6s_3 + 4s_4 + 2s_5 > \frac{8}{11} + \frac{3}{10} = \frac{113}{110} > 1.$$

This proves all signs in (35) and $\gamma = 1/2 - B_3 > 15/64$.

Expanding G_1 from (38) using the s_i definitions gives its displayed radical expression. The bounds (80) then imply

$$260(608276) + 96(1360147) + 420(223606) = 382640392,$$

$$G_1 > \frac{382640392 - 382500000}{13200000} = \frac{17549}{1650000} > 0.$$

The same direct expansion gives (74). Its lower certificate is

$$180(608276) + 60(1360147) + 420(223606) = 285013020,$$

$$G_4 > \frac{285013020 - 285100000}{13200000} = -\frac{4349}{660000}.$$

Finally the common-denominator calculation in (76) has numerator $-4349 \cdot 16 + 5 \cdot 20625 = -69584 + 103125 = 33541$ and denominator 10560000. This completes all arithmetic certificates.

11 Positive distances, arbitrary sets, and the witness

We finish the proof of Theorem 1.1. If eleven distinct points had common distance $\delta > 0$, replacing every coordinate by x_k/δ would give eleven unit-equilateral points, since

$$\sum_{k=1}^{5} \left| \frac{x_k}{\delta} - \frac{y_k}{\delta} \right| = \frac{1}{\delta} \sum_{k=1}^{5} |x_k - y_k| .$$

This is impossible by the preceding sections. Any finite equilateral set with more than ten elements contains eleven distinct elements. An infinite set also contains eleven distinct elements: choose successively a point outside the previously chosen finite subset. Thus an arbitrary equilateral set is finite and has at most ten elements. This argument places no finiteness assumption on the original set.

For the lower bound take

$$S_{10} = \{\tfrac{1}{2}e_i, -\tfrac{1}{2}e_i : 1 \le i \le 5\}.$$

These are ten distinct points. Opposite points on the same coordinate axis have distance one. Points on different axes have two nonzero coordinate differences of magnitude $1/2$, again giving distance one. This proves the asserted witness and $e(\ell_1^5) = 10$.

12 Formalization and declaration correspondence

The accompanying formal development uses Lean and mathlib [2, 5]. Points are functions from a finite set of coordinate indices to $\mathbb{R}$, and their distance is defined literally as the finite sum of absolute coordinate differences. Its eleven-point input is an arbitrary function $\{1, \ldots, 11\} \to \mathbb{R}^5$. The unit-equilateral predicate quantifies over every pair of distinct labels, so it includes all 55 unordered pairs.

The matrix presentation in Section 2 corresponds to scalar cut energies in the formal sources: on a zero-sum vector, a scalar column energy is the sum of its nonnegative gaps multiplied by squared prefix sums. The formal total-energy and total-trace identities are the identities (3)–(5) evaluated in this form. Thus the matrix notation in the paper abbreviates exactly the ordered scalar computation used by the formal proof.

The following table identifies the main correspondence. All names are in the namespace `Kusner5`; entries with a further namespace retain it in the table.

Mathematical statement	Formal declaration
Exact all-pair energy and trace identities	`unit_equilateral_actual_totalEnergy`; `unit_equilateral_five_totalTrace`
The physical range bound, Proposition 3.2	`actual_coordinate_range_le_five_sixths`
At least four wide coordinates and exact shifted cross energy	`Profiles.at_least_four_wide`; `Profiles.shifted_crossEnergy`
At least two isolated edges and all five possible count cases	`ForestMatching.original_at_least_two_isolated`; `ForestMatching.original_exhaustive_counts`
The two cases with exactly two isolated edges	`ForestMatching.no_four_wide_two_isolated`; `ForestMatching.no_five_wide_two_isolated`
Four-edge matching and three-isolated-edge path	`FourMatchingAssembly.no_four_matching`; `PathAssembly.no_five_wide_three_isolated`
Quartet cut totals, original overlap, and all four cross equations	`quartet_equal_cut_totals`; `quartet_original_cooccurrence`; `quartet_original_cross_equation`
Actual projection and positive active-face projection	`exists_original_quartet_projection`; `exists_original_quartet_face_projection`
Physical slot classification and the same seven-label lower cycle	`original_quartet_positive_slot_classification`; `original_projected_seven_cycle_lower`
The two physical boundary tours and cyclic upper bound	`exists_original_113_section_tour`; `exists_original_122_section_tour`; `QuartetGeometry.seven_cycle_of_closed_path`
Arbitrary original quartet width, Proposition 8.1	`original_quartet_singleton_ge_one_twenty_fourth`; `original_quartet_width_ge_thirty_seven_twenty_fourths`
Five actual matching rows and the selected quartet slack	`MatchingAssembly.original_five_matching_energy_lower`; `MatchingAssembly.matching_original_slack`
Eleven-point unit-distance exclusion, with the literal expanded input	`no_eleven_unit`; `no_eleven_unit_explicit`
Positive-distance normalization and finite-array bound	`no_eleven_positive_distance`; `equilateral_array_card_le_ten`
Arbitrary-set finiteness and cardinality bound	`equilateral_set_finite_card_le_ten`
The ten-point witness and complete exact result	`exists_equilateral_ten`; `kusner_five_exact`

In particular, the fully expanded eleven-point theorem has the mathematical type

$$\forall x : \{1, \ldots, 11\} \to \mathbb{R}^5, \quad \neg \left(\forall i \neq j,\ \sum_{k=1}^{5} |x_{i,k} - x_{j,k}| = 1 \right).$$

The final exact theorem combines the arbitrary-set, arbitrary-positive- distance upper bound with the explicit finite ten-point lower witness. The arbitrary-quartet lower bound is proved from the original configuration before it is supplied to the final matching argument; it is not an additional hypothesis of the final theorem.

The development reuses mathlib's real-number, finite-sum, finite-set, sorting, and square-root results. The project declarations and their composition are kernel-checked, and the transitive axiom inspection of the central and final declarations yields only the standard foundations `propext`, `Classical.choice`, and `Quot.sound`. All numerical certificates above are represented by exact arithmetic proofs. No unproved mathematical axiom or admitted proof is used. This verification statement concerns the supplied formal development with its declared library dependencies.

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